Chunghwan Lee

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Education

Hanyang University (HYU) Seoul, Korea
M.S., Dept. of Electronic Engineering (*GPA: 4.31/4.5*) *Mar.* 2022 – Feb. 2024
Thesis: *Paired-Point Lifting for Privacy-Preserving Visual Localization*
Advisor: Prof. Je Hyeong Hong

Hanyang University (HYU) Seoul, Korea
B.S., Dept. of Electronic Engineering (*GPA: 3.82/4.5*) *Mar.* 2015 – Feb. 2022
Thesis: *Deepfake Detection Using Supervised Temporal Feature Extraction Model and LSTM*
Advisor: Prof. Kijung Yoon

Skills

Programming proficient in: Python (including libraries such as OpenCV, Open3D, COLMAP); familiar with: C/C++, Java, MATLAB
Deep learning framework proficient in: PyTorch, Keras and Tensorflow (including visualization and debugging tools such as Tensorboard and Wandb)

Publications (International)

1.**"Paired-Point Lifting for Enhanced Privacy-Preserving Visual Localization"**, **Chunghwan Lee**, Jaihoon Kim, Chanhyuk Yun, Je Hyeong Hong, *The IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023*

2. **"Efficient Privacy-Preserving Visual Localization Using 3D Ray Clouds"**, Heejoon Moon, **Chunghwan Lee**, Je Hyeong Hong, *The IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024*

Patents

(*Korea*) "Localization method and apparatus using line cloud map data, line cloud map data generation method", Je Hyeong Hong, **Chunghwan Lee**, Chanhyuk Yun, *Registration No. 10-2023-0008666 (applicant: HYU)* *Jan. 2023*

Research Projects

NaverLabs, Research Internship, Spatial AI team Seongnam, Gyeonggi-do, Korea
Generalizable 3D Gaussian Splatting *Mar. 2024 – Present*
- Developing generalizable Gaussian generator for sparse-view 3D scene reconstruction.

HY-Vision Lab, Hanyang University (HYU) Seoul, Korea
Enhancing Privacy-Preserving Visual Localization *Mar. 2022 – Feb. 2024*
- Developed privacy preserving 3D line cloud generation method which draw lines by pairing two points, named paired-point lifting(PPL).
- Secured scene recovery attack that intrudes privacy without compromising camera pose re-localization accuracy.

Deep Learning-based Sketched Face Recognition *Oct. 2021 – Feb.2022*
- Developed human sketched face recognition model by fine-tuning the ARKFACE and conducting an augmentation on sketch data with CycleGAN.

NIAI Lab, Hanyang University (HYU) Seoul, Korea
Deepfake Video Detection *Feb. 2021 – Sep. 2021*
- Developed classification model discerning deepfake videos by training the model to extract the abrupt features with PCA-based loss.

ICSP Lab, Hanyang University (HYU) Seoul, Korea
Image-based Weather Classification and Restoration *Jun. 2020 – Oct. 2020*
- Devised one-for-all model that processes from classification to reconstruction from bad weather artifacts; Derain, Desnow, Dehaze.

Honors	**Third Place**, IC-PBL+ Graduate Championship 2022-2, IC-PBL Teaching and Learning Center, HYU, Seoul, Korea	*2022*
	Excellence Award, Undergraduate Hanyang Academic, HYU, Seoul, Korea	*2022*
	Excellence Award, The Korean Institute of Broadcast and Media Engineers Fall Conference 2021, The KIBME, Seoul, Korea	*2021*
Scholarships	**HY-in Scholarships**, Graduate Hanyang University, Seoul, Korea	*2022 – present* $ 12,200
	Dept. E.E. Scholarship, Undergraduate Hanyang University, Seoul, Korea	*2015 – 2016* $ 14,500
Additional	**Paper reviewer - ICIP 2022-2023, WACV 2023-2024, CVPR 2024**	